

Bootstrap 5.3

Add BS to project:

There are two main ways to add Bootstrap 5.3 to your project:

1. Use a CDN: A CDN is a content delivery network that hosts Bootstrap 5.3 files on servers all over the world. This means that your users can download the Bootstrap 5.3 files from a server that is close to them, which can improve the performance of your website.

To add Bootstrap 5.3 to your project using a CDN, simply add the following code to the header of your HTML document:

```
<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet"                                integrity="sha384-
rwcTc8fdN4dZLvMQm8YZwaPRWD1omZ1kVSFq30f3zJ38UL09MYwohXupnAQd3a0Mn"
crossorigin="anonymous">

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"
integrity="sha384-
w7skcQhTbaC09v35jedmOtRVw08ped90Ad50bYvGUm/uWuQCwS0tx9R09s/HVqZ6"
crossorigin="anonymous"></script>
```

2. Download and install Bootstrap 5.3: You can also download and install Bootstrap 5.3 on your own server. This gives you more control over the Bootstrap 5.3 files, but it also means that you will need to keep them updated yourself.

To download and install Bootstrap 5.3, visit the Bootstrap website and click on the "Download" button. Extract the downloaded files to a directory on your server.

Then, add the following code to the header of your HTML document:

```
<link href="/path/to/bootstrap/css/bootstrap.min.css" rel="stylesheet">
<script src="/path/to/bootstrap/js/bootstrap.bundle.min.js"></script>
```

Once you have added Bootstrap 5.3 to your project, you can start using Bootstrap 5.3 components and styles in your HTML document.

Here are some additional tips for adding Bootstrap 5.3 to your project:

- If you are using a CDN, make sure that you are using the latest version of Bootstrap 5.3. You can check the Bootstrap website for the latest version number.
- If you are downloading and installing Bootstrap 5.3 on your own server, make sure to keep the Bootstrap 5.3 files updated. You can check the Bootstrap website for updates.
- You can use the Bootstrap documentation to learn more about how to use Bootstrap 5.3 components and styles.

Typography:

Typography in Bootstrap 5.3 is based on the following principles:

- **Usefulness:** The typography should be clear, easy to read, and efficient.
- **Flexibility:** The typography should be flexible enough to be used in a variety of contexts and devices.
- **Consistency:** The typography should be consistent throughout the website.

Bootstrap 5.3 uses a system of font sizes, line heights, and font weights to create a typographic hierarchy that is easy to read and understand. The following are the most important aspects of typography in Bootstrap 5.3:

- **Font size:** The base font size in Bootstrap 5.3 is 1rem, which is equal to 16px. This is a good size for most body text. Bootstrap 5.3 also provides a set of heading sizes that are based on the base font size.
- **Line height:** The line height in Bootstrap 5.3 is 1.5. This means that there is 1.5 times the space between lines of text. This helps to make text more readable.
- **Font weight:** The font weight in Bootstrap 5.3 is 400 for body text and 700 for headings. This creates a clear distinction between body text and headings.

Bootstrap 5.3 also provides a number of other typographic features, such as:

- **Text alignment:** Bootstrap 5.3 provides a number of ways to align text, including left, right, center, and justify.
- **Text color:** Bootstrap 5.3 provides a number of ways to color text, including primary, secondary, success, danger, warning, and info.
- **Text decoration:** Bootstrap 5.3 provides a number of ways to decorate text, such as underline, overline, strikethrough, and italic.

To use Bootstrap 5.3 typography, you can use the following CSS classes:

- **.fs-1 to .fs-6:** These classes set the font size of the text.
- **.lh-1 to .lh-6:** These classes set the line height of the text.
- **.fw-light to .fw-bold:** These classes set the font weight of the text.
- **.text-left to .text-right:** These classes align the text to the left or right.
- **.text-primary to .text-info:** These classes color the text to the primary, secondary, success, danger, warning, or info color.
- **.text-decoration-none to .text-decoration-underline:** These classes decorate the text with underline, overline, strikethrough, or italic.

You can also use the Bootstrap 5.3 typography utility classes to quickly change the font size, line height, font weight, text alignment, text color, or text decoration of text. For example, to set the font size of a paragraph to 24px, you can use the .fs-2 class.

Bootstrap 5.3 typography is a powerful tool that can help you to create a website that is easy to read and understand. By using the Bootstrap 5.3 typography system, you can create a typographic hierarchy that is clear, consistent, and flexible.

Image

Images in Bootstrap 5.3 are made responsive with the .img-fluid class. This applies max-width: 100%; and height: auto; to the image so that it scales with the parent element.

To add an image to your Bootstrap 5.3 website, simply use the element and add the .img-fluid class. For example:

```

```

You can also use Bootstrap 5.3 to align images and add borders to them. For example, to center an image, you can use the .mx-auto class. To add a border to an image, you can use the .img-thumbnail class.

Table

Tables in Bootstrap 5.3 are made responsive with the .table class. This applies a variety of CSS styles to the table, including border-collapse: collapse; and width: 100%;.

To add a table to your Bootstrap 5.3 website, simply use the <table> element and add the .table class. For example:

HTML

```
<table class="table">
  <thead>
    <tr>
      <th>#</th>
      <th>First Name</th>
      <th>Last Name</th>
      <th>Username</th>
    </tr>
  </thead>
  <tbody>
    <tr>
```

```

        <td>1</td>
        <td>John</td>
        <td>Doe</td>
        <td>@johndoe</td>
    </tr>
    <tr>
        <td>2</td>
        <td>Jane</td>
        <td>Doe</td>
        <td>@janedoe</td>
    </tr>
</tbody>
</table>

```

You can also use Bootstrap 5.3 to add borders to tables and style the cells in the table. For example, to add a border to a table, you can use the `.table-bordered` class. To style the cells in a table, you can use the `.table-striped`, `.table-active`, and `.table-hover` classes.

Figure

The figure component in Bootstrap 5.3 is used to display images and captions together. To add a figure to your Bootstrap 5.3 website, simply use the `<figure>` element.

For example, the following code will display an image with a caption:

```

<figure class="figure">
    
    <figcaption class="figure-caption">This is my image.</figcaption>
</figure>

```

You can also use Bootstrap 5.3 to align figures and style the captions. For example, to center a figure, you can use the `.mx-auto` class. To style the caption of a figure, you can use the `.figure-caption` class.

Conclusion

Images, tables, and figures are all important components of a website. Bootstrap 5.3 provides a variety of CSS classes that can be used to style these components and make them responsive. By using the Bootstrap 5.3 image, table, and figure components, you can create a website that is visually appealing and easy to use.

Grid

Bootstrap is a popular front-end framework that provides a grid system to create responsive layouts for websites and web applications. The grid system in Bootstrap is based on a 12-column layout. It allows you to build complex and flexible designs by using a combination of rows and columns. Here's a brief explanation of the essential classes used in Bootstrap's grid system:

1. **container**: This class is used to create a fixed-width container for the content. It provides padding on the left and right sides of the container.
2. **container-fluid**: This class creates a full-width container that spans the entire width of the viewport. It does not have any padding on the left and right sides.
3. **row**: The **row** class is used to create a horizontal group of columns. It must be used within a **container** or **container-fluid** to properly align the columns.
4. **col-{breakpoint}-{number}**: These classes define the columns for different screen sizes. The **{breakpoint}** can be **xs** (extra small devices), **sm** (small devices), **md** (medium devices), **lg** (large devices), or **xl** (extra-large devices), and the **{number}** represents the number of columns the element should span.

For example, **col-md-6** creates a column that spans 6 out of the 12 columns for medium-sized devices.

5. **offset-{breakpoint}-{number}**: This class is used to offset columns from the left. It creates empty spaces to the left of the columns based on the specified number of columns.

For example, **offset-lg-3** creates an offset of 3 columns for large devices.

By combining these classes, you can create responsive grid layouts in Bootstrap that adjust according to the screen size. The flexibility of the Bootstrap grid system allows developers to design visually appealing and functional websites that work seamlessly across various devices and screen sizes.

Code Example:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1" />
    <link
      href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
      rel="stylesheet"
    />
    <link
      href="https://getbootstrap.com/docs/5.3/assets/css/docs.css"
      rel="stylesheet"
    />
    <title>Bootstrap Example</title>
    <script
      src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"
    ></script>
  </head>
  <body class="p-3 m-0 border-0 m-0 border-0 bd-example-row">
    <!-- Example Code -->
    <div class="container text-center">
      <div class="row g-2">
        <div class="col-12 p-2">col</div>
        <div class="col-12">col</div>
        <div class="col-12">col</div>
        <div class="col-12">col</div>
      </div>
      <div class="row">
        <div class="col-8">col-8</div>
        <div class="col-4">col-4</div>
        <div class="col-12">col8</div>
      </div>
    </div>
  </body>
</html>
```

```
        <!-- End Example Code -->
    </body>
</html>
```


Container and Utility

Utility and container styling in Bootstrap 5.3 are two important ways to customize the look and feel of your website.

Utility styling is used to quickly and easily apply common CSS styles to elements on your page. For example, you can use utility classes to add borders, colors, and padding to elements.

Container styling is used to create a layout for your website. Containers are typically used to wrap around the main content of your page and provide a consistent width and padding.

To use utility and container styling in Bootstrap 5.3, simply add the appropriate CSS classes to your HTML elements. For example, to add a border to a paragraph element, you would add the `.border` class. To create a container for your main content, you would add the `.container` class.

Here are some examples of how to use utility and container styling in Bootstrap 5.3:

```
<p class="border">This paragraph has a border.</p>

<div class="container">

  <h1>This is my website!</h1>

  <p>This is the main content of my website.</p>

</div>
```

In addition to the basic utility and container classes, Bootstrap 5.3 also provides a number of other utility and container classes that you can use to customize the look and feel of your website. For more information, please see the Bootstrap documentation.

Here are some of the most commonly used utility and container classes in Bootstrap 5.3:

Utility classes:

- **.border:** Adds a border to an element.
- **.color:** Sets the color of an element.
- **.padding:** Adds padding to an element.
- **.margin:** Adds margin to an element.
- **.float:** Floats an element.
- **.position:** Positions an element.
- **.width:** Sets the width of an element.
- **.height:** Sets the height of an element.

Container classes:

- **.container:** Creates a container for the main content of your page.
- **.container-fluid:** Creates a fluid container that will fill the entire width of the browser window.
- **.row:** Creates a row of columns within a container.
- **.col:** Creates a column within a row.

You can combine utility and container classes to create custom styles for your website. For example, you could use the `.border` and `.padding` classes to add a border and padding to a container. Or, you could use the `.col-md-6` class to create a column that is half the width of the container on medium-sized screens and above.

Utility and container styling are powerful tools that can help you to create a website that is both visually appealing and easy to use. By using utility and container styling, you can quickly and easily customize the look and feel of your website without having to write any custom CSS.

Form

Below is an explanation of each of the Bootstrap form components you mentioned:

1. **Form control:** A fundamental building block for creating various form elements such as input fields, text areas, checkboxes, radio buttons, and more. It allows users to input or select data and interact with the web application.
2. **Select:** A form control that allows users to select options from a dropdown list. It provides a convenient way to choose from a predefined set of options and is commonly used for selecting single or multiple values.
3. **Checks & radios:** Form controls used for collecting binary or multiple-choice input from users. Checks are used for checkboxes, allowing users to select multiple options, while radios are used for radio buttons, allowing users to select only one option from a set of choices.
4. **Range:** A form control that enables users to select a value from a range or slider. It is commonly used for selecting numerical values within a specific range, such as selecting a value from a range of numbers or setting a volume control.
5. **Input group:** A versatile form component that combines an input field with other elements such as buttons, dropdowns, or text, providing users with additional context or options related to the input field. Input groups help enhance the usability and functionality of input fields.
6. **Floating labels:** A form design technique where the placeholder text moves to the top of the input field when the user focuses on the field or enters data. It helps improve the visibility of the input labels and provides a better user experience by keeping the labels visible during data entry.
7. **Layout:** The arrangement and organization of form elements and controls within a web page or application. Bootstrap provides a grid system and various layout components that allow developers to create responsive and user-friendly forms with different arrangements, alignments, and styles.
8. **Validation:** The process of checking and verifying user input to ensure that it meets specific criteria or requirements. Bootstrap provides validation styles and JavaScript plugins that help developers create interactive and user-friendly forms with real-time validation feedback, such as displaying error messages or highlighting invalid input fields.

Below is an example of a simple HTML form that incorporates various Bootstrap form components. The code is well-commented to explain the purpose of each form component.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Bootstrap Form Components Example</title>
    <!-- Bootstrap CSS -->
    <link
      href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.cs
s"
      rel="stylesheet"
      integrity="sha384-
T3c6CoIi6uLrA9TneNEoa7RxnatzjcDSCmG1MXxSR1GAsXEV/Dwwykc2MPK8M2HN"
      crossorigin="anonymous"
    />
  </head>
  <body class="bg-light-subtle">
    <div class="container-fluid">
      <h1 class="text-center">Bootstrap Forms</h1>
      <form>
        <!-- Form Control -->
        <div class="ipGroup w-50 bg-warning p-2 rounded">
          <h2>Simple Input</h2>
          <!-- Layout -->
          <div class="row mt-1">
            <div class="col">
              <label for="exampleInputEmail1" class="form-label fw-bold">
                >First Name</label>
              <input
                type="text"
                class="form-control"
                placeholder="First name">
            </div>
          </div>
        </div>
      </form>
    </div>
  </body>
</html>
```

```

        />
    </div>
    <div class="col">
        <label for="exampleInputEmail1" class="form-label fw-bold"
            >Last Name</label>
        >
        <input type="text" class="form-control" placeholder="Last name" />
    </div>
</div>
<div class="mb-3">
    <label for="exampleInputEmail1" class="form-label fw-bold"
        >Email address</label>
    >
    <input
        type="email"
        class="form-control"
        id="exampleInputEmail1"
        aria-describedby="emailHelp"
    />
</div>
<div class="mb-3">
    <label for="exampleInputPassword1" class="form-label fw-bold"
        >Password</label>
    >
    <input
        type="password"
        class="form-control"
        id="exampleInputPassword1"
    />
</div>
<button type="submit" class="btn btn-success">Submit</button>
</div>
<div class="formOtherComp w-50 bg-info mt-1 p-3 rounded">
    <h2>Form Other Component 1</h2>

```

```

<!-- Select -->
<div class="form-group">
  <label for="exampleFormControlSelect1">Example select</label>
  <select class="form-control w-50" id="exampleFormControlSelect1">
    <option>1</option>
    <option>2</option>
    <option>3</option>
    <option>4</option>
    <option>5</option>
  </select>
</div>

<!-- Checks & Radios -->
<div class="form-check">
  <h2>Checkbox Example</h2>
  <div class="check1">
    <input
      class="form-check-input"
      type="checkbox"
      value=""
      id="defaultCheck1"
    />
    <label class="form-check-label" for="defaultCheck1">
      checkbox 1
    </label>
  </div>
  <div class="check1">
    <input
      class="form-check-input"
      type="checkbox"
      value=""
      id="defaultCheck1"
    />
    <label class="form-check-label" for="defaultCheck1">
      checkbox 2
    </label>
  </div>
</div>

```

```

        </label>
    </div>
    <div class="check1">
        <input
            class="form-check-input"
            type="checkbox"
            value=""
            id="defaultCheck1"
        />
        <label class="form-check-label" for="defaultCheck1">
            checkbox 3
        </label>
    </div>
</div>
<div class="form-check">
    <h2>Radio Example</h2>
    <div class="radio1">
        <input
            class="form-check-input"
            type="radio"
            name="exampleRadios"
            id="exampleRadios1"
            value="option1"
            checked
        />
        <label class="form-check-label" for="exampleRadios1">
            Radio option 1
        </label>
    </div>
    <div class="radio2">
        <input
            class="form-check-input"
            type="radio"
            name="exampleRadios"

```

```

        id="exampleRadios1"
        value="option1"
        checked
    />
    <label class="form-check-label" for="exampleRadios1">
        Radio Option 2
    </label>
</div>
<div class="radio3">
    <input
        class="form-check-input"
        type="radio"
        name="exampleRadios"
        id="exampleRadios1"
        value="option1"
        checked
    />
    <label class="form-check-label" for="exampleRadios1">
        Radio Option 3
    </label>
</div>
</div>
<!-- Range -->
<div class="form-group">
    <label for="formControlRange">Example Range input</label>
    <input
        type="range"
        class="form-control-range"
        id="formControlRange"
    />
</div>
</div>

<div class="formOtherCont w-50 bg-success p-3 mt-2 rounded">

```



```

<!-- Input Group -->
<h2>Form Other Component 2</h2>
<div class="input-group mb-3">
  <div class="input-group-prepend">
    <span class="input-group-text" id="basic-addon1">@</span>
  </div>
  <input
    type="text"
    class="form-control"
    placeholder="Username"
    aria-label="Username"
    aria-describedby="basic-addon1"
  />
</div>
<!-- Floating Labels -->
<div class="form-floating">
  <input
    type="email"
    class="form-control"
    id="floatingInput"
    placeholder="name@example.com"
  />
  <label for="floatingInput">Email address</label>
</div>
</div>
</form>
</div>

<script
  src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.m
in.js"
  integrity="sha384-
C6RzsynM9kWDrMNeT87bh95OGNyZPhcTNXj1NW7RuBCsyN/o0jlpcV8Qyq46cDfL"
  crossorigin="anonymous"
></script>

```

</body>

</html>

Bootstrap Components

Bootstrap is a popular front-end framework that provides a wide range of components to build responsive and user-friendly web applications. Here is an explanation of some of the common Bootstrap components you mentioned:

1. **Accordion:** A collapsible set of content panels that allows you to toggle the display of content sections. Clicking on one section collapses the others.
2. **Alerts:** Dismissible components for conveying important information to the user. They come in different styles such as success, warning, info, and danger, each with its own color scheme.
3. **Badge:** Small, informative labels that can be added to links, buttons, or other components to provide additional context or information.
4. **Breadcrumb:** A navigation aid that helps users visualize their location within a website or application's hierarchy. It displays the current page's location relative to the homepage.
5. **Buttons:** Interactive elements that users can click to perform actions. Bootstrap provides a variety of button styles, including default, primary, secondary, success, warning, danger, and more.
6. **Button group:** A collection of buttons grouped together, often used for related actions or options. Button groups can be customized to include dropdown menus and other interactive elements.
7. **Card:** A versatile content container that houses various types of content, such as text, images, and links. Cards can be customized and styled to create visually appealing components.
8. **Carousel:** A slideshow component for cycling through a series of content items, such as images or text. It provides options for controlling the speed, navigation, and appearance of the slideshow.
9. **Close button:** A small button typically used to dismiss alerts, modals, or other components. Clicking the close button hides or removes the associated content.
10. **Collapse:** A component that allows you to hide or show content. It can be used to create collapsible content areas, such as collapsible panels or dropdown menus.
11. **Dropdowns:** Interactive components that enable users to select options from a list. Dropdowns are often used in navigation menus or form elements to provide a list of selectable choices.
12. **List group:** A versatile component for displaying lists of content. It can be used to create various types of lists, including simple lists, linked lists, and lists with contextual classes for highlighting certain items.

Below is an example of a simple HTML page that incorporates various Bootstrap components. The code is well-commented to explain each component's purpose.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Components Example</title>
  <!-- Bootstrap CSS -->
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">
</head>
<body>
  <div class="container">
    <!-- Accordion -->
    <div id="accordionExample">
      <!-- Accordion Card 1 -->
      <div class="card">
        <div class="card-header" id="headingOne">
          <h2 class="mb-0">
            <button class="btn btn-link" type="button" data-
toggle="collapse" data-target="#collapseOne" aria-expanded="true" aria-
controls="collapseOne">
              Accordion Item #1
            </button>
          </h2>
        </div>
        <div id="collapseOne" class="collapse show" aria-
labelledby="headingOne" data-parent="#accordionExample">
          <div class="card-body">
            Content for Accordion Item #1
          </div>
        </div>
      </div>
    </div>
  </div>
</body>
</html>
```

```

        <!-- Other Accordion Cards... -->
    </div>
    <!-- Alerts -->
    <div class="alert alert-success" role="alert">
        A simple success alert—check it out!
    </div>
    <!-- Badge -->
    <button type="button" class="btn btn-primary">
        Notifications <span class="badge badge-light">4</span>
    </button>
    <!-- Breadcrumb -->
    <nav aria-label="breadcrumb">
        <ol class="breadcrumb">
            <li class="breadcrumb-item"><a href="#">Home</a></li>
            <li class="breadcrumb-item"><a href="#">Library</a></li>
            <li class="breadcrumb-item active" aria-current="page">Data</li>
        </ol>
    </nav>
    <!-- Buttons -->
    <button type="button" class="btn btn-primary">Primary</button>
    <button type="button" class="btn btn-secondary">Secondary</button>
    <!-- Button Group -->
    <div class="btn-group">
        <button type="button" class="btn btn-danger">Action</button>
        <button type="button" class="btn btn-danger dropdown-toggle dropdown-
toggle-split" data-toggle="dropdown" aria-haspopup="true" aria-expanded="false">
            <span class="sr-only">Toggle Dropdown</span>
        </button>
        <div class="dropdown-menu">
            <a class="dropdown-item" href="#">Action</a>
            <a class="dropdown-item" href="#">Another action</a>
            <a class="dropdown-item" href="#">Something else here</a>
        </div>
    </div>

```

```

<!-- Card -->
<div class="card" style="width: 18rem;">
  
  <div class="card-body">
    <h5 class="card-title">Card title</h5>
    <p class="card-text">Some quick example text to build on the card
title and make up the bulk of the card's content.</p>
    <a href="#" class="btn btn-primary">Go somewhere</a>
  </div>
</div>
<!-- Carousel -->
<div id="carouselExampleSlidesOnly" class="carousel slide" data-
ride="carousel">
  <div class="carousel-inner">
    <div class="carousel-item active">
      
    </div>
    <div class="carousel-item">
      
    </div>
    <!-- More carousel items... -->
  </div>
</div>
<!-- Close Button -->
<button type="button" class="close" aria-label="Close">
  <span aria-hidden="true">&times;</span>
</button>
<!-- Collapse -->
<p>
  <a class="btn btn-primary" data-toggle="collapse"
href="#collapseExample" role="button" aria-expanded="false" aria-
controls="collapseExample">
    Link with href
  </a>
</p>

```

```

<div class="collapse" id="collapseExample">
  <div class="card card-body">
    Content for Collapse Component
  </div>
</div>

<!-- Dropdowns -->
<div class="dropdown">
  <button class="btn btn-secondary dropdown-toggle" type="button"
id="dropdownMenuButton" data-toggle="dropdown" aria-haspopup="true" aria-
expanded="false">

    Dropdown button
  </button>
  <div class="dropdown-menu" aria-labelledby="dropdownMenuButton">
    <a class="dropdown-item" href="#">Action</a>
    <a class="dropdown-item" href="#">Another action</a>
    <a class="dropdown-item" href="#">Something else here</a>
  </div>
</div>

<!-- List Group -->
<ul class="list-group">
  <li class="list-group-item">Cras justo odio</li>
  <li class="list-group-item">Dapibus ac facilisis in</li>
  <li class="list-group-item">Morbi leo risus</li>
  <li class="list-group-item">Porta ac consectetur ac</li>
</ul>

</div>

<!-- Bootstrap JS (optional) -->
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>

<script
src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"></
script>

<script
src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script
>

</body>

```

</html>

Components Continued:

1. **List group:** A versatile component for displaying a series of content, such as a list of items or links. It can be styled to create various types of lists, including basic lists, linked lists, and lists with contextual classes for highlighting certain items.
2. **Modal:** A dialog box that appears on top of the main content to provide additional information or functionality. Modals can be used for alerts, notifications, forms, or other interactive components that require user input.
3. **Navbar:** A navigation bar component that provides a consistent and easy-to-use way to navigate through a website or application. It typically contains links, buttons, and other elements for site navigation and branding.
4. **Navs & tabs:** Components for creating navigation menus with tabs. They allow users to switch between different sections of content within the same page, improving the overall user experience and navigation.
5. **Offcanvas:** A component for creating a hidden sidebar or content panel that can be revealed by clicking on a button or trigger. Offcanvas is often used for displaying additional content, navigation menus, or options without cluttering the main interface.
6. **Pagination:** A component for dividing content into multiple pages, allowing users to navigate through long lists or sets of data. Pagination enhances the usability of a website by improving content organization and accessibility.
7. **Placeholders:** Styling elements used as temporary representations of content that will be replaced later. They are commonly used for demonstrating the layout and structure of a web page before the actual content is available.
8. **Popovers:** Interactive components that provide additional information or context when users hover over or click on an element. Popovers are often used to display tooltips, additional details, or related content in a small overlay box.
9. **Progress:** A visual indicator that displays the progress of a task or operation. The progress component is often used to show the completion status of a process, such as file uploads, form submissions, or data loading.
10. **Scrollspy:** A component that automatically updates the navigation links based on the scroll position. Scrollspy is commonly used in long pages or content-heavy websites to indicate the user's current position and provide a smooth navigation experience.
11. **Spinners:** Animated loading indicators that show the progression of a task or operation. Spinners are commonly used to indicate that a process is ongoing and to provide visual feedback to users while they wait for content to load.

12. **Toasts:** Non-intrusive, temporary notifications that appear on the screen to provide feedback or alerts to the user. Toasts are often used to display success messages, error notifications, or other important updates without disrupting the user's workflow.
13. **Tooltips:** Interactive components that provide additional information or context when users hover over an element. Tooltips are commonly used to display descriptive text, hints, or explanatory content related to specific UI elements.

Here's an example of a simple HTML page that incorporates various Bootstrap components. The code is well-commented to explain the purpose of each component.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Components Example</title>
  <!-- Bootstrap CSS -->
  <link href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
  <div class="container">
    <!-- List Group -->
    <ul class="list-group">
      <li class="list-group-item">Item 1</li>
      <li class="list-group-item">Item 2</li>
      <li class="list-group-item">Item 3</li>
    </ul>
    <!-- Modal -->
    <button type="button" class="btn btn-primary" data-toggle="modal" data-target="#myModal">
      Open Modal
    </button>
    <div class="modal" id="myModal">
      <div class="modal-dialog">
        <div class="modal-content">
          <div class="modal-header">
```

```

        <h4 class="modal-title">Modal Title</h4>
        <button type="button" class="close" data-
dismiss="modal">&times;</button>
    </div>
    <div class="modal-body">
        Modal Content
    </div>
    <div class="modal-footer">
        <button type="button" class="btn btn-secondary" data-
dismiss="modal">Close</button>
    </div>
</div>
</div>
</div>
<!-- Navbar -->
<nav class="navbar navbar-expand-lg navbar-light bg-light">
    <a class="navbar-brand" href="#">Navbar</a>
    <button class="navbar-toggler" type="button" data-toggle="collapse"
data-target="#navbarNav" aria-controls="navbarNav" aria-expanded="false" aria-
label="Toggle navigation">
        <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarNav">
        <ul class="navbar-nav">
            <li class="nav-item active">
                <a class="nav-link" href="#">Home <span class="sr-
only">(current)</span></a>
            </li>
            <li class="nav-item">
                <a class="nav-link" href="#">Features</a>
            </li>
            <li class="nav-item">
                <a class="nav-link" href="#">Pricing</a>
            </li>
        </ul>
    </div>
</div>

```

```

</nav>

<!-- Navs & Tabs -->
<ul class="nav nav-tabs">
  <li class="nav-item">
    <a class="nav-link active" href="#">Tab 1</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Tab 2</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Tab 3</a>
  </li>
</ul>

<!-- Offcanvas -->
<button class="btn btn-primary" type="button" data-
toggle="offcanvas">Toggle Offcanvas</button>

<div class="offcanvas offcanvas-start" tabindex="-1" id="offcanvasExample"
aria-labelledby="offcanvasExampleLabel">
  <div class="offcanvas-header">
    <h5 class="offcanvas-title"
id="offcanvasExampleLabel">Offcanvas</h5>
    <button type="button" class="btn-close text-reset" data-bs-
dismiss="offcanvas" aria-label="Close"></button>
  </div>
  <div class="offcanvas-body">
    Offcanvas Content
  </div>
</div>

<!-- Pagination -->
<nav aria-label="Page navigation example">
  <ul class="pagination">
    <li class="page-item"><a class="page-link"
href="#">Previous</a></li>
    <li class="page-item"><a class="page-link" href="#">1</a></li>
    <li class="page-item"><a class="page-link" href="#">2</a></li>
    <li class="page-item"><a class="page-link" href="#">3</a></li>
  </ul>
</nav>

```

```

        <li class="page-item"><a class="page-link" href="#">Next</a></li>
    </ul>
</nav>

<!-- Placeholders -->
<div class="p-3 mb-2 bg-light text-dark">Placeholder Content</div>

<!-- Popovers -->
<button type="button" class="btn btn-secondary" data-toggle="popover"
title="Popover Title" data-content="Popover Content">Popover Button</button>

<!-- Progress -->
<div class="progress">
    <div class="progress-bar" role="progressbar" style="width: 25%;" aria-
valuenow="25" aria-valuemin="0" aria-valuemax="100"></div>
</div>

<!-- Scrollspy -->
<body data-spy="scroll" data-target="#navbar-example">

<!-- Spinners -->
<div class="spinner-border text-primary" role="status">
    <span class="visually-hidden">Loading...</span>
</div>

<!-- Toasts -->
<div class="toast" role="alert" aria-live="assertive" aria-atomic="true">
    <div class="toast-header">
        <strong class="me-auto">Bootstrap</strong>
        <small>11 mins ago</small>
        <button type="button" class="btn-close" data-bs-dismiss="toast"
aria-label="Close"></button>
    </div>
    <div class="toast-body">
        Hello, world! This is a toast message.
    </div>
</div>

<!-- Tooltips -->
<button type="button" class="btn btn-secondary" data-toggle="tooltip" data-
placement="top" title="Tooltip Message">Tooltip Button</button>

</div>

```

```
<!-- Bootstrap JS (optional) -->

<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>

<script
src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"></
script>

<script
src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script
>

<!-- Bootstrap Popover & Tooltip Initialization -->

<script>

    $(function () {

        $('[data-toggle="popover"]').popover();

        $('[data-toggle="tooltip"]').tooltip();

    });

</script>
</body>
</html>
```

